Kinematics Practice Questions — Easy to Medium (50 Questions)

Distance/Displacement/Speed/Velocity · Averaging Trap · Instantaneous Velocity · Momentarily at Rest · Acceleration & Signs · Equations of Motion · nth Second

A. DISTANCE, DISPLACEMENT, SPEED & VELOCITY

Q1. A man walks 3 km east, then 4 km north. Find the total distance travelled and the displacement.

Q2. A body moves along a square track of side 10 m and completes 1.5 rounds. Find the distance covered and the displacement.

Q3. A car travels 40 km in 1 hour, then 60 km in 1.5 hours, in the same direction. Find its average speed and average velocity.

Q4. A man walks 8 km east in 2 hours, then 6 km west in 1 hour. Find his average speed and average velocity.

Q5. A body travels one-third of its total distance at 10 m/s and the remaining two-thirds at 15 m/s. Find the average speed.

Q6. A particle covers 5 m in the first 2 seconds and 15 m in the next 2 seconds (same direction throughout). Find its average velocity over the full 4 seconds.

Q7. A train travels 100 km north in 2 hours, then returns 100 km south in 2 hours along the same route. Find the average speed and average velocity for the whole trip.

Q8. A swimmer covers 500 m downstream in 5 minutes, then swims back 500 m upstream in 10 minutes. Find the average speed and average velocity for the whole trip.

B. THE AVERAGING TRAP

Q9. A car covers the first half of a distance at 30 km/h and the second half at 60 km/h. Find the average speed.

Q10. A car covers the first half of the TIME of its journey at 30 km/h and the second half at 60 km/h. Find the average speed.

Q11. A cyclist covers half the distance of a journey at 12 km/h and the other half at 18 km/h. Find the average speed.

Q12. A train travels for half the total time at 50 km/h and the other half at 70 km/h. Find the average speed.

Q13. A bus covers half the distance of a trip at 20 km/h and the other half at 80 km/h. Find the average speed.

Q14. A car travels for half the total time at 20 km/h and the other half at 80 km/h. Find the average speed.

C. INSTANTANEOUS VELOCITY

Q15. The position of a particle is $x = 5t^2 + 3t$ (SI). Find its instantaneous velocity at $t = 4$ s.

Q16. $x = 2t^3 - 4t$ (SI). Find the instantaneous velocity at $t = 2$ s.

Q17. $x = t^2 - 6t + 5$ (SI). Find the velocity at $t = 0$ s and at $t = 3$ s. Is the particle speeding up or slowing down between these times?

Q18. A particle moves according to $x = 4t^3$ (SI). Find its instantaneous speed at $t = 1$ s.

Q19. For $x = 3t^2 + 2$ (SI), find the average velocity between $t = 1$ s and $t = 3$ s, and compare it with the instantaneous velocity at $t = 2$ s.

Q20. $x = 7t - t^2$ (SI). Find the velocity at $t = 2$ s and at $t = 5$ s. At what time is the particle momentarily at rest?

Q21. $x = 5t^2 - 2t + 1$ (SI). Find the instantaneous velocity at $t = 3$ s.

Q22. $x = t^3 + 2t$ (SI). Find the average velocity from $t = 0$ to $t = 2$ s, and the instantaneous velocity at $t = 1$ s. Are they equal?

D. MOMENTARILY AT REST

Q23. $x = t^3 - 3t^2$ (SI). Find $v(t)$ and the instant(s) when the particle is momentarily at rest.

Q24. $x = t^3 - 12t$ (SI). Find the time(s) when the particle is momentarily at rest.

Q25. $x = 2t^3 - 9t^2 + 12t$ (SI). Find all instants when the particle is momentarily at rest.

Q26. A ball's height is $h = 20t - 5t^2$ (SI). Find the time at which its velocity is zero.

Q27. $x = t^3 - 6t^2 + 9t$ (SI). Find the instant(s) when momentarily at rest.

Q28. $x = t^3 - 9t^2 + 24t$ (SI). Find the instant(s) when momentarily at rest.

E. ACCELERATION & SIGN CONVENTIONS

Q29. A particle has velocity $v = -8$ m/s and acceleration $a = -3$ m/s². Is it speeding up or slowing down? Explain using signs.

Q30. A particle has $v = +6$ m/s and $a = -4$ m/s². Is it speeding up or slowing down?

Q31. A car moving at 200 km/h maintains a constant speed for 10 minutes. What is its acceleration? Explain why.

Q32. A particle has $v = -5$ m/s and $a = +2$ m/s². Is it speeding up or slowing down? What will eventually happen to its direction of motion?

Q33. A particle has $v = +10$ m/s and $a = +5$ m/s². Is it speeding up or slowing down?

Q34. A particle has $v = 0$ and $a = -9.8$ m/s² at a given instant. Is the particle accelerating? What will happen to it in the next instant?

F. THE THREE EQUATIONS OF MOTION

Q35. A scooter accelerates uniformly from rest at 3 m/s^2 for 5 s. Find its final velocity and the distance covered.

Q36. A car moving at 18 m/s decelerates uniformly and comes to rest after covering 81 m. Find the deceleration and the time taken.

Q37. A ball moves along a straight track with $u = 12$ m/s and uniform deceleration $a = -2$ m/s^2. Find its velocity at $t = 3$ s and the distance covered in that time.

Q38. A bike accelerates uniformly from 5 m/s to 17 m/s in 6 s. Find the acceleration and the distance covered.

Q39. A train travels at 25 m/s for 10 s, then decelerates uniformly at 5 m/s^2 until it stops. Find the total distance covered during the entire motion.

Q40. A car starts from rest and accelerates uniformly at 4 m/s^2 for 6 s. Find its final velocity and the distance covered.

Q41. A body has initial velocity 20 m/s and uniform deceleration 4 m/s^2. Find the time it takes to stop and the distance covered before stopping.

Q42. A particle has $u = 10$ m/s, $v = 30$ m/s, and uniform acceleration 4 m/s^2. Find the time taken and the distance covered.

Q43. A car decelerates uniformly from 20 m/s to 5 m/s over 15 s. Find the deceleration and the distance covered.

Q44. A body starts with $u = 6$ m/s and uniform acceleration 2 m/s^2. Find its velocity and displacement after 4 s.

G. DISPLACEMENT IN THE NTH SECOND

Q45. A body starts from rest with acceleration 4 m/s^2. Find (a) the distance covered in 4 seconds, and (b) the distance covered in the 4th second. State the units of each answer clearly.

Q46. A particle starts from rest with uniform acceleration 6 m/s^2. Find the displacement in the 3rd second.

Q47. A car moving with uniform acceleration covers 10 m in the 3rd second and 16 m in the 5th second. Find its initial velocity and acceleration.

Q48. A body starts from rest with uniform acceleration 5 m/s^2. Find the ratio of distances covered in the 1st, 2nd, and 3rd seconds.

Q49. A particle has $u = 2$ m/s and uniform acceleration 3 m/s^2. Find its displacement in the 4th second.

Q50. A body starts from rest with uniform acceleration 8 m/s^2. Find the distance covered in the 5th second.

ANSWER KEY

SECTION A

Q1. Dist=7km, Disp=5km
Q2. Dist=60m, Disp=10root2
approx 14.1m
Q3. Speed=40km/h
Velocity=40km/h
Q4. Speed~4.67km/h
Velocity~0.67km/h(E)
Q5. Speed~12.86 m/s
Q6. Velocity=5 m/s
Q7. Speed=50km/h
Velocity=0
Q8. Speed~1.11 m/s
Velocity=0

SECTION B

Q9. 40 km/h
Q10. 45 km/h
Q11. 14.4 km/h
Q12. 60 km/h
Q13. 32 km/h
Q14. 50 km/h

SECTION C

Q15. v=43 m/s
Q16. v=20 m/s
Q17. v(0)=-6, v(3)=0
slowing down
Q18. v=12 m/s
Q19. avg=12, inst=12
equal
Q20. v(2)=3, v(5)=-3
rest at t=3.5s
Q21. v=28 m/s
Q22. avg=6, inst=5
not equal

SECTION D

Q23. v=3t2-6t
rest: t=0, 2s
Q24. t=2s
Q25. t=1s, 2s
Q26. t=2s
Q27. t=1s, 3s
Q28. t=2s, 4s

SECTION E

Q29. same signs
speeding up
Q30. opposite signs
slowing down
Q31. a=0, v constant
Q32. opposite signs
slowing down,
will reverse
Q33. same signs
speeding up
Q34. still accelerating
will start moving
in -ve direction

SECTION F

Q35. v=15 m/s, s=37.5m
Q36. a=-2 m/s2, t=9s
Q37. v=6 m/s, s=27m
Q38. a=2 m/s2, s=66m
Q39. total s=312.5m
Q40. v=24 m/s, s=72m
Q41. t=5s, s=50m
Q42. t=5s, s=100m
Q43. a=-1 m/s2
s=187.5m
Q44. v=14 m/s, s=40m

SECTION G

Q45. (a) 32 m
(b) 14 m
Q46. s_3rd=15 m
Q47. u=2.5 m/s
a=3 m/s2
Q48. Ratio=1:3:5
Q49. s_4th=12.5 m
Q50. s_5th=36 m